ARCHITECTURE OF 8086 MICROPROCESSOR

✅ General Information

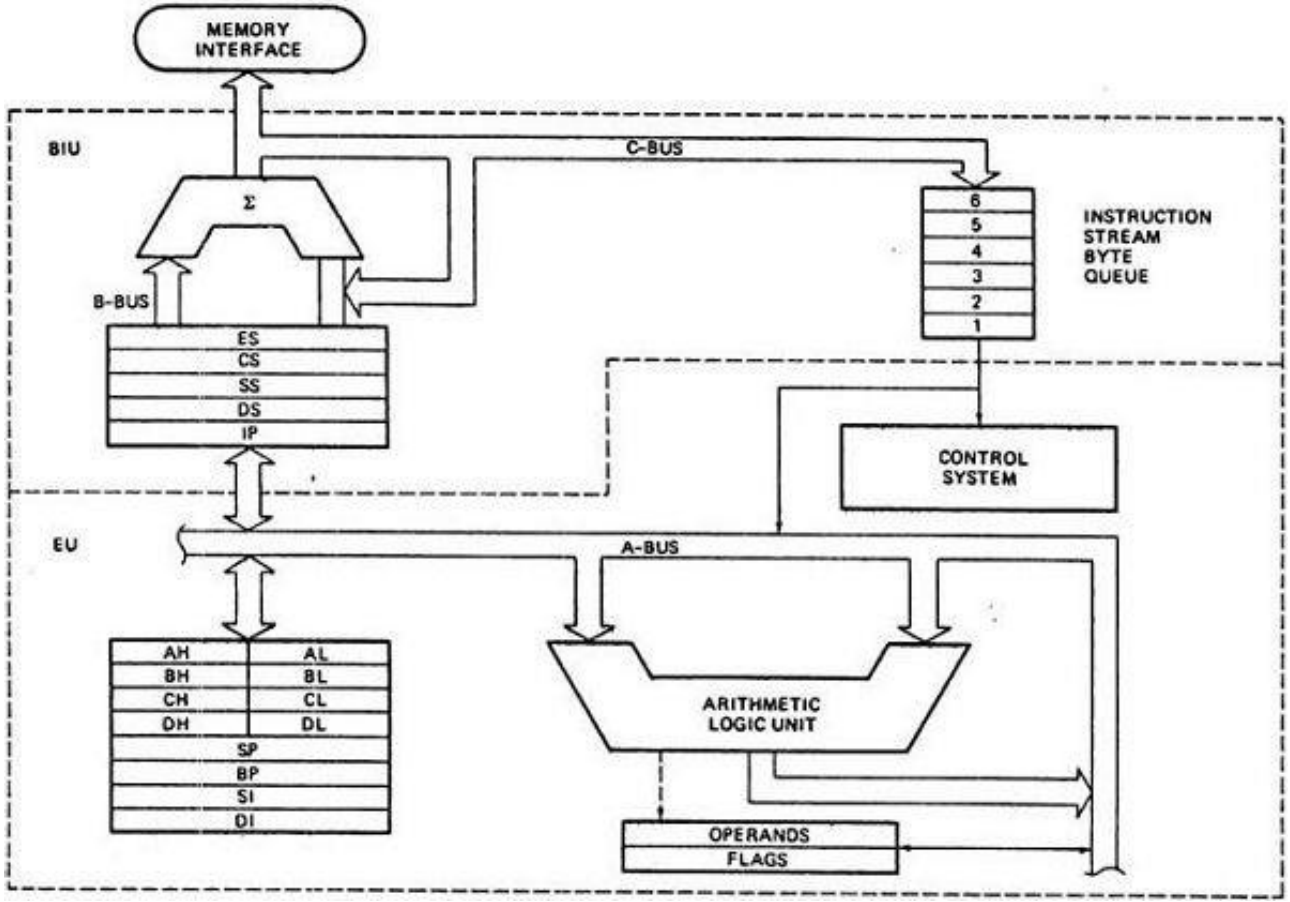
1. **Launch Year**: 1978
2. **Bit Size**: First **16-bit** microprocessor.
3. **Transistors**: Contains **29,000** transistors.
4. **Package Type**: Available in a **40-pin Dual-Inline Package (DIP)**.
5. **Improvement**: Major speed and architecture improvement over **Intel 8085**.

✅ Available Versions

- **8086** – 5 MHz
- **8086-2** – 8 MHz
- **8086-1** – 10 MHz

- **16-bit Processor**: Data bus width is 16 bits.
- **20-bit Address Bus**: Can access **2^{20} = 1 MB** memory (from 00000H to FFFFFH).
- **Prefetch Queue**: Can **fetch 6 instruction bytes** ahead to speed up execution (pipelining).
- **Clock Requirement**: Needs a **single-phase clock with 33% duty cycle**.
- **Power Supply**: Operates on a **+5V power supply**.

ARCHITECTURE OF 8086 MICROPROCESSOR

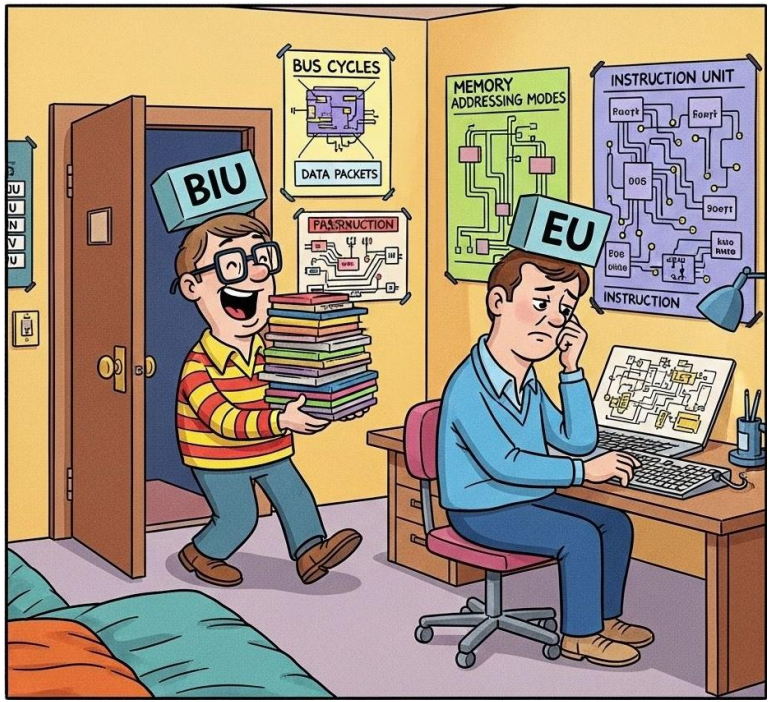


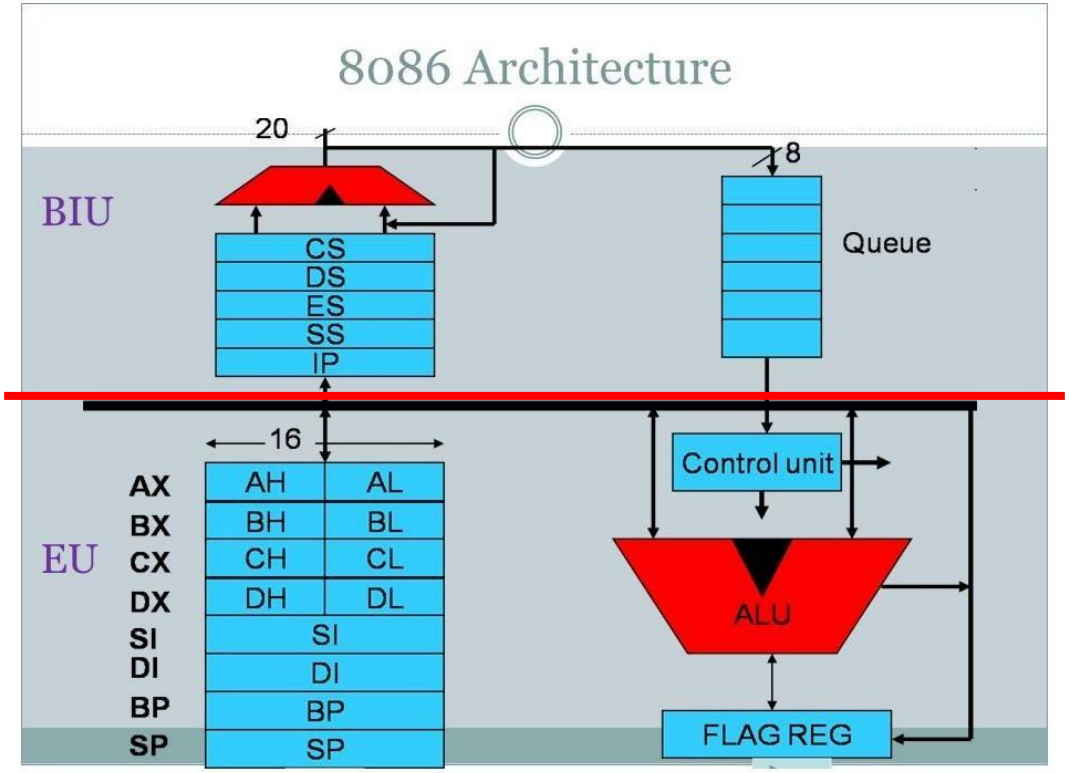
BEFORE

	1	2	3	4	5	6
I_1	F	D	E			
I_2				F	D	E

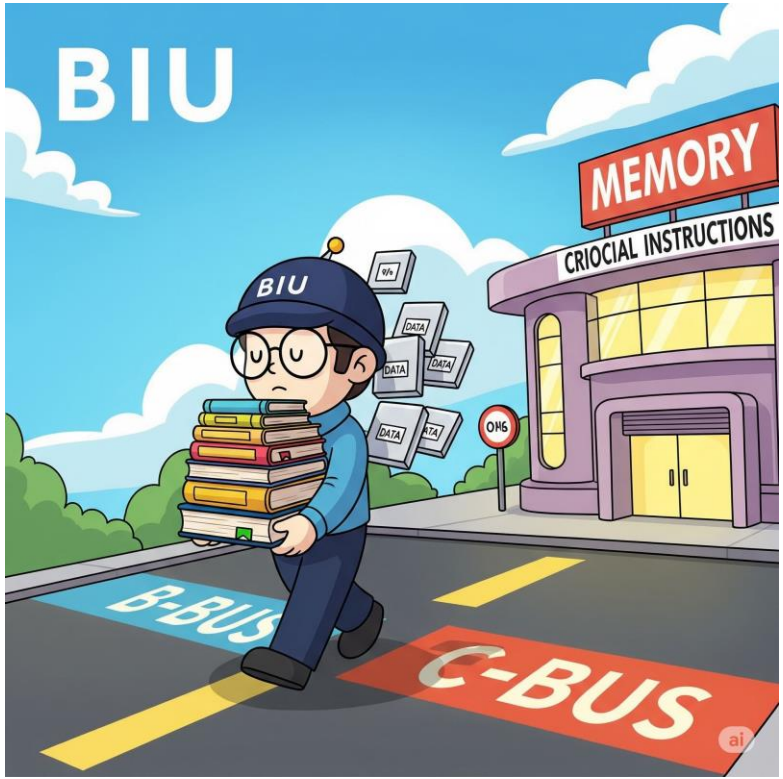
AFTER

	1	2	3	4	5	6
I_1	F	D	E			
I_2		F	D	E		





Bus Interface Unit (BIU) & Execution Unit (EU)



Bus Interface Unit (BIU):

The Bus Interface Unit (BIU) is responsible for communicating with memory and I/O devices. It acts like the messenger of the CPU, managing data flow between the processor and external systems. It handles:

- **Fetching instructions from memory.**
- **Reading/writing data to and from memory or I/O ports.**
- **Calculating physical addresses using segment registers and offset.**
- **Queuing instructions using a 6-byte prefetch queue to speed up execution.**
- **Controlling the system bus (address and data bus).**

Execution Unit (EU):

The Execution Unit (EU) is the part that actually executes the instructions fetched by the BIU. It includes:

- **Arithmetic Logic Unit (ALU) – performs calculations (add, sub, etc.).**
- **Control unit – decodes instructions and controls execution.**
- **Registers – stores temporary data, like AX, BX, CX, DX, etc.**
- **Flag register – holds status flags (zero, carry, sign, etc.).**

ARCHITECTURE OF 8086 MICROPROCESSOR

1. A-Bus (Address Bus)

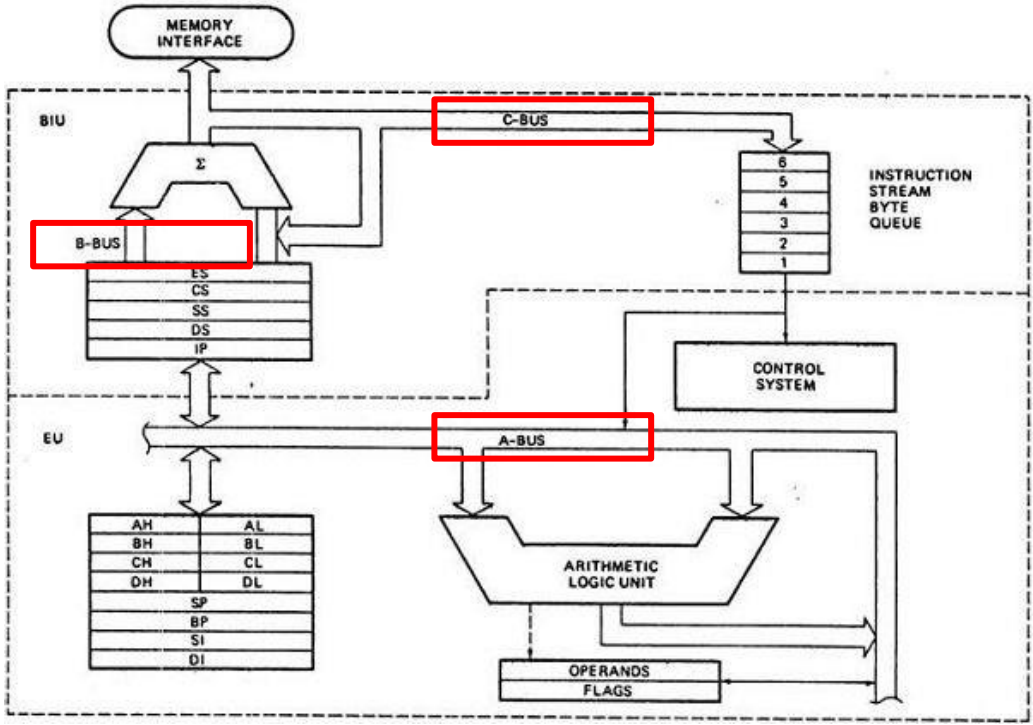
- Carries address information.
- Used internally for effective address calculation.
- Transfers addresses between registers and address generation logic.

2. B-Bus (Data Bus)

- Carries operand/data to the ALU.
- Connects general-purpose registers to the ALU.
- Used during arithmetic and logical operations.

3. C-Bus (Control Bus)

- Carries control signals.
- Tells internal components what operation to perform (e.g., read/write, ALU commands).
- Controls data flow and operations within the processor.



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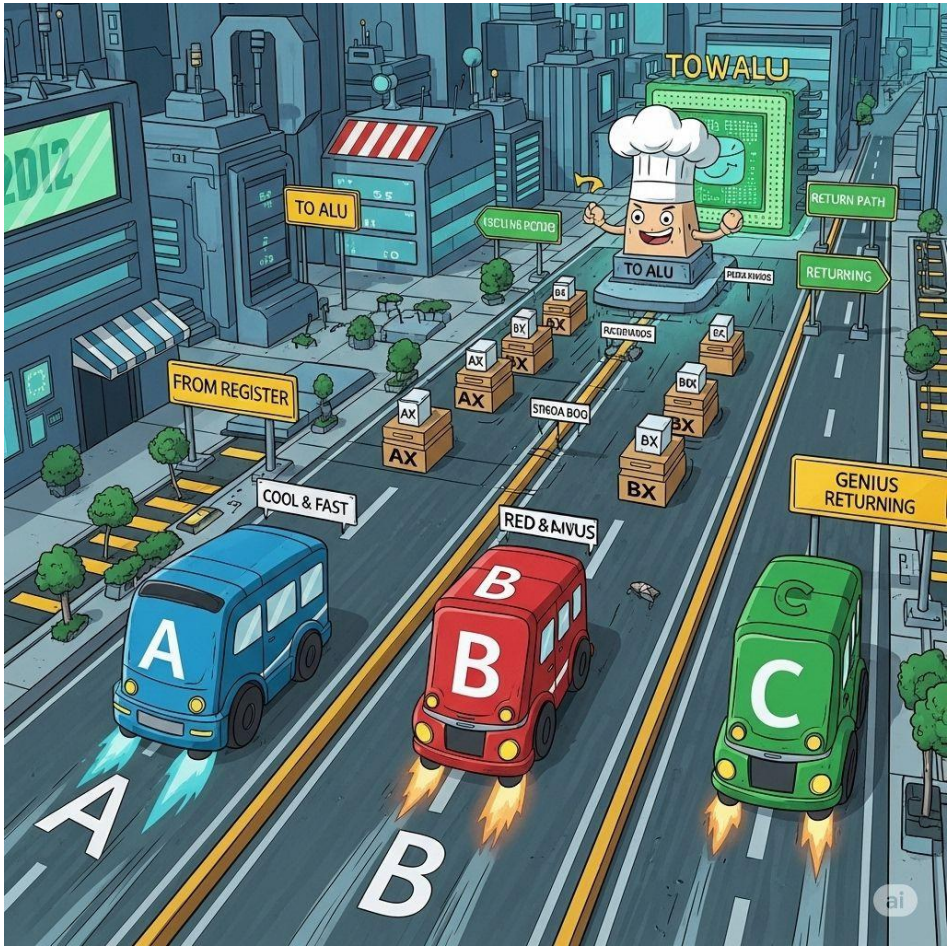
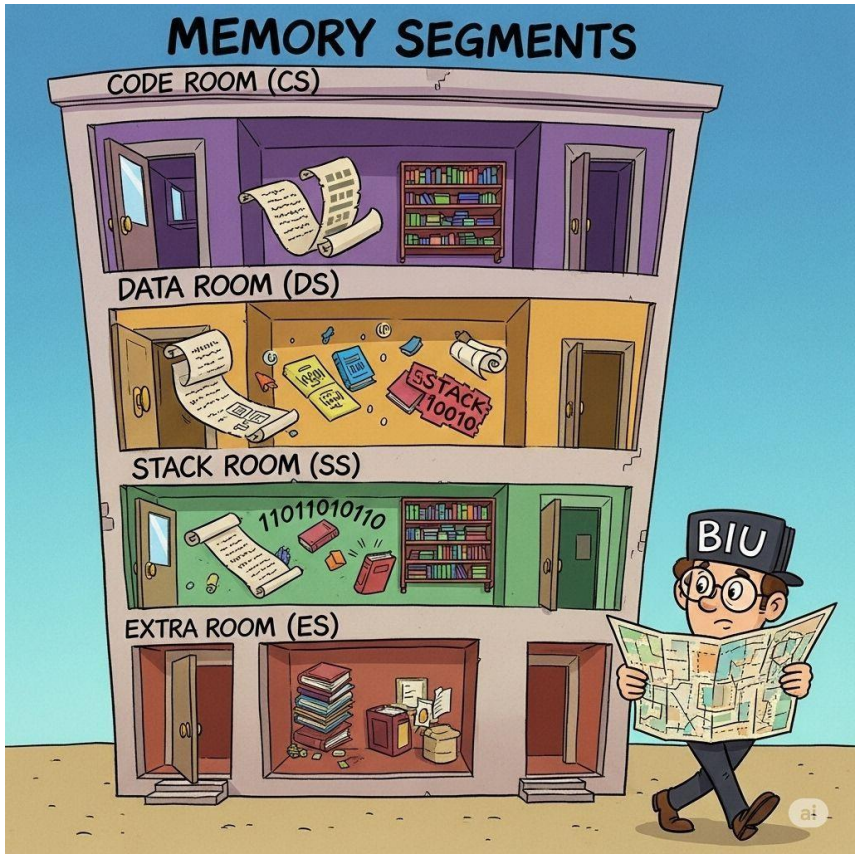
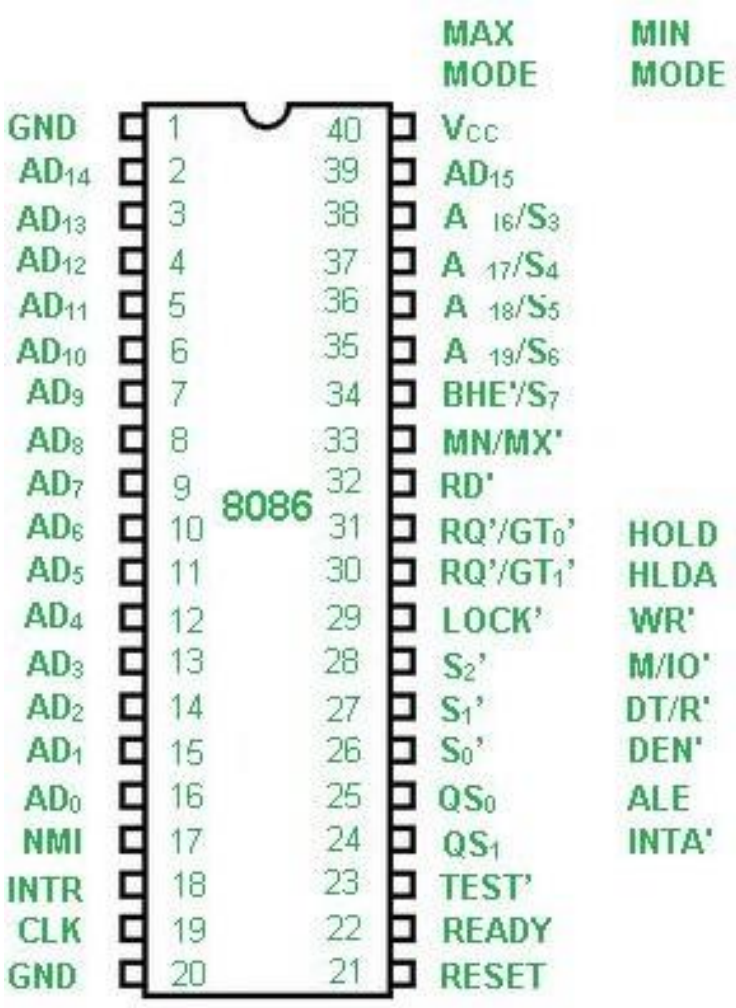


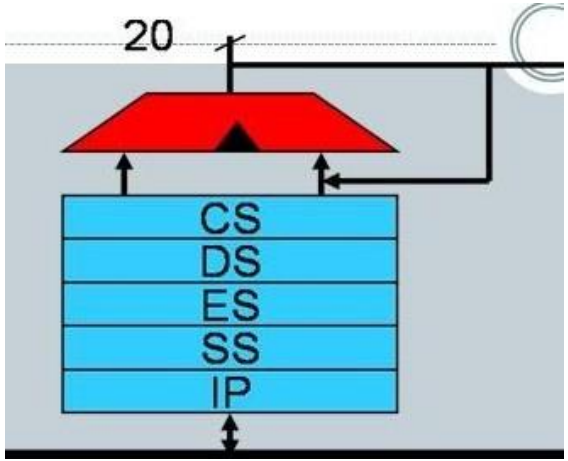
BIU বাবুর স্টাক বাড়ি সফর



🧰 Advantages of Segmentation

- Logical organization of code, data, and stack.
- Security and protection (in more advanced CPUs).
- Enables modular program design.
- Efficient memory utilization.


MAX MODE
MIN MODE
GND 1
AD14 2
AD13 3
AD12 4
AD11 5
AD10 6
AD9 7
AD8 8
AD7 9
AD6 10
AD5 11
AD4 12
AD3 13
AD2 14
AD1 15
AD0 16
NMI 17
INTR 18
CLK 19
GND 20
8086
40 VCC
39 AD15
38 A16/S3
37 A17/S4
36 A18/S5
35 A19/S6
34 BHE'/S7
33 MN/MX'
32 RD'
31 RQ'/GT0' HOLD
30 RQ'/GT1' HLDA
29 LOCK' WR'
28 S2' M/IO'
27 S1' DT/R'
26 S0' DEN'
25 QS0 ALE
24 QS1 INTA'
23 TEST'
22 READY
21 RESET



S4 (A17)	S3 (A16)	Segment
0	0	Extra Segment (ES)
0	1	Stack Segment (SS)
1	0	Code Segment (CS)
1	1	Data Segment (DS)

CS (Code Segment):

- Works with the **IP (Instruction Pointer)** to fetch instructions.
- `Physical Address = CS × 10H + IP`

DS (Data Segment):

- Works with general-purpose registers (e.g., AX, BX, etc.)
- Example: `MOV AX, [BX]` uses DS as the segment base by default.

SS (Stack Segment):

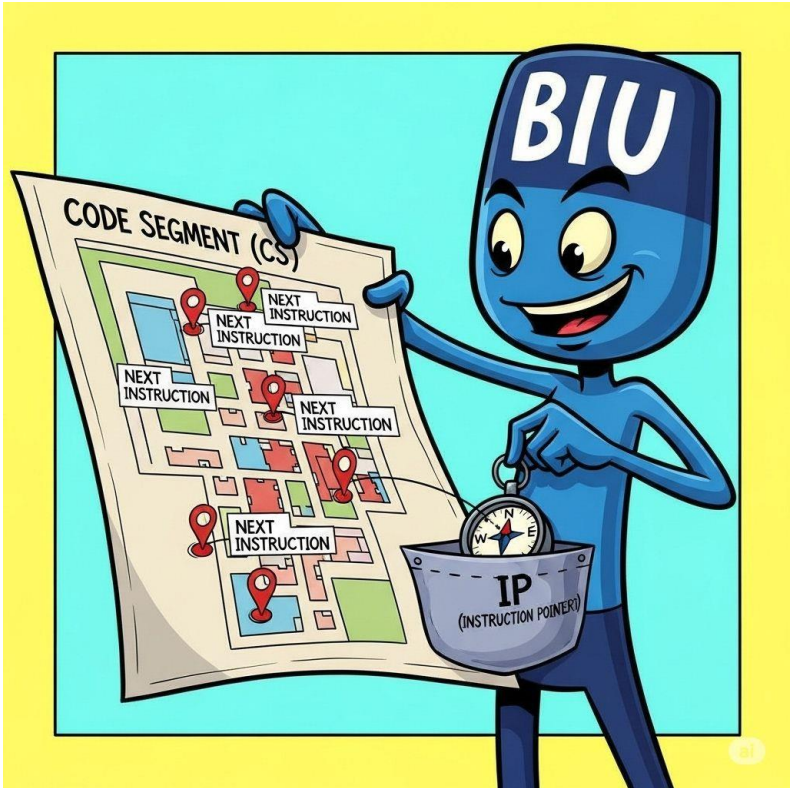
- Works with **SP (Stack Pointer)** and **BP (Base Pointer)**.
- Used during **PUSH, POP, CALL, RET**.

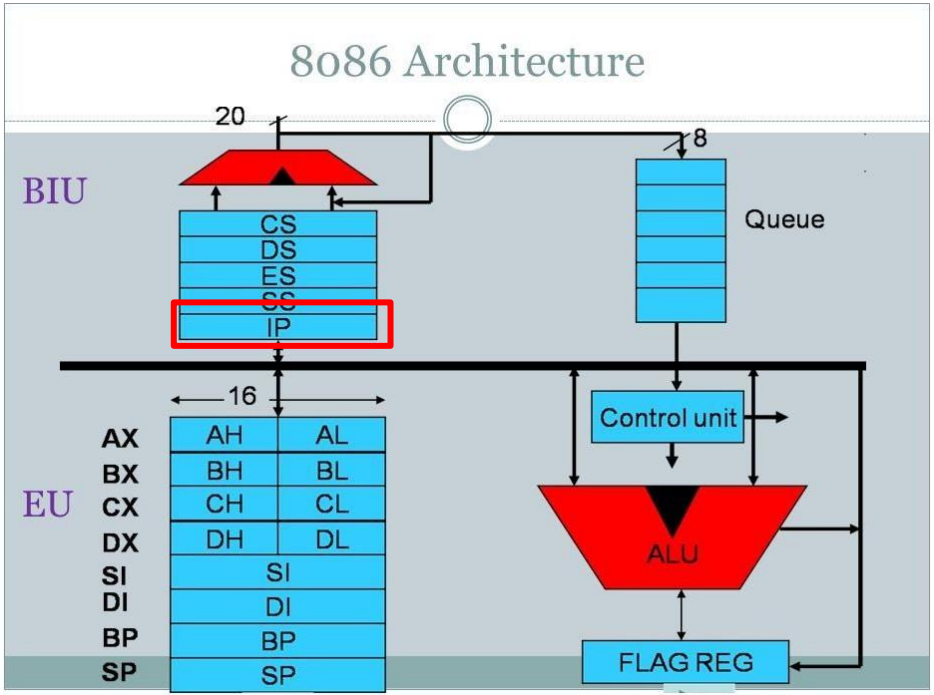
ES (Extra Segment):

- Used in **string operations**, especially with **DI (Destination Index)**.
- Example: `MOVSB`, `MOVSW` use ES:DI and DS:SI.

IP (Instruction Pointer) is a 16-bit register that holds the offset of the next instruction within the Code Segment (CS).

The physical address is calculated as: **(CS × 10H) + IP**.





What is Physical Address?

- The **actual 20-bit memory address** used by the 8086 microprocessor to access memory.
- Calculated using **Segment Register** and **Offset**.

📌 **Formula:**

🧮 **Physical Address = (Segment × 10H) + Offset**

- `Segment`: The base address (e.g., CS, DS, SS, ES)
- `10H` = 16 (to shift segment left by 4 bits)
- `Offset`: Distance from the start of the segment

Example:

- **CS = 1234H , IP = 0020H**

Physical Address = (1234H × 10H) + 0020H
= 12340H + 0020H
= 12360H

Offset

Offset is the distance (in bytes) from the beginning of a memory segment to a specific data or instruction location within that segment.

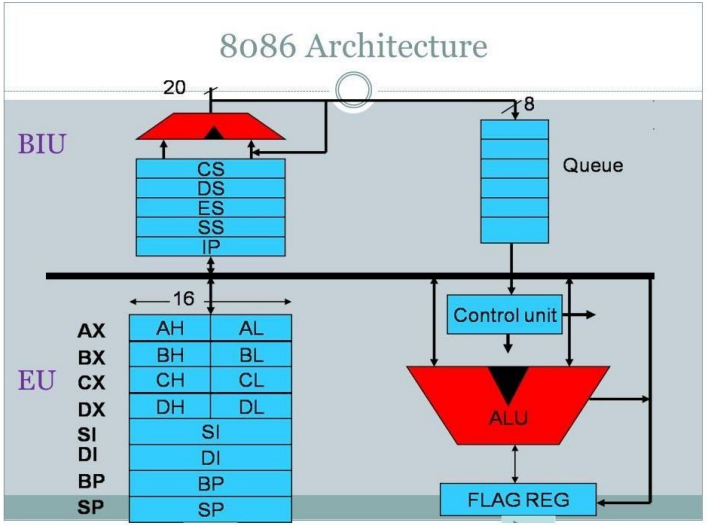
It is a **16-bit value** used in combination with a **segment register** to calculate the **physical memory address** in the 8086 microprocessor.

🧮 **Physical Address = (Segment × 10H) + Offset**

236 km (segment)

58.1 km (offset)

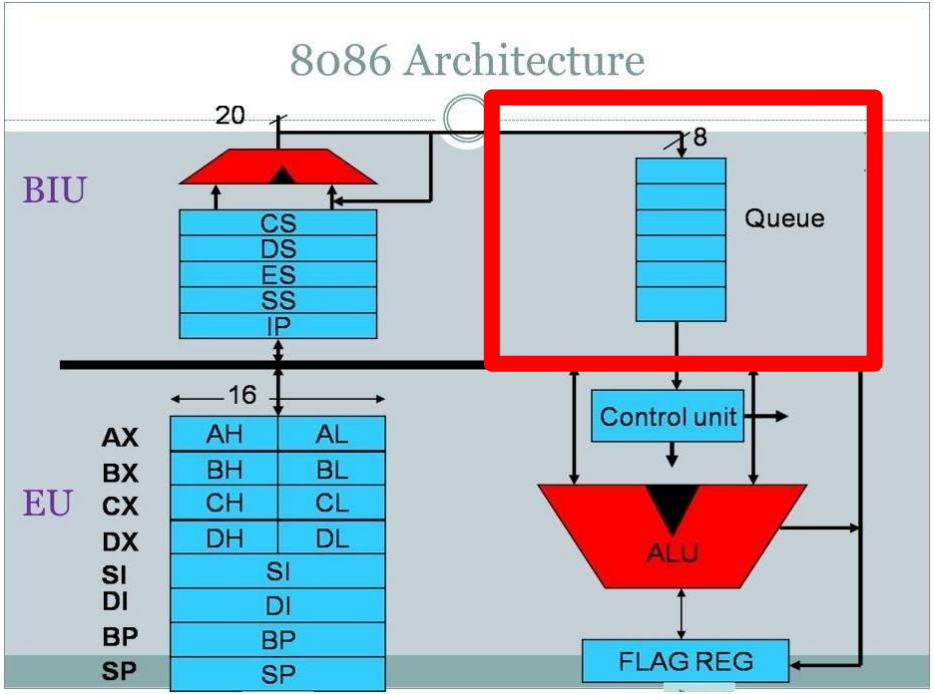
Segment Register	Used For	Offset Comes From
CS (Code Segment)	Executing instructions	**IP (Instruction Pointer)**
DS (Data Segment)	Accessing data	AX, BX, CX, DX, SI, DI, BP (typically SI/DI)
SS (Stack Segment)	Stack operations	**SP (Stack Pointer)** / BP
ES (Extra Segment)	String operations, extra data	**DI (Destination Index)**



BiU এর রান্নাঘর

Instruction Queue :

An **Instruction Queue** is a temporary storage in the CPU where instructions are pre-fetched and held before execution to improve processing speed and efficiency.



- **AX (Accumulator Register)**

Used for arithmetic, logic, and I/O operations. Often the default register in many instructions.

- **BX (Base Register)**

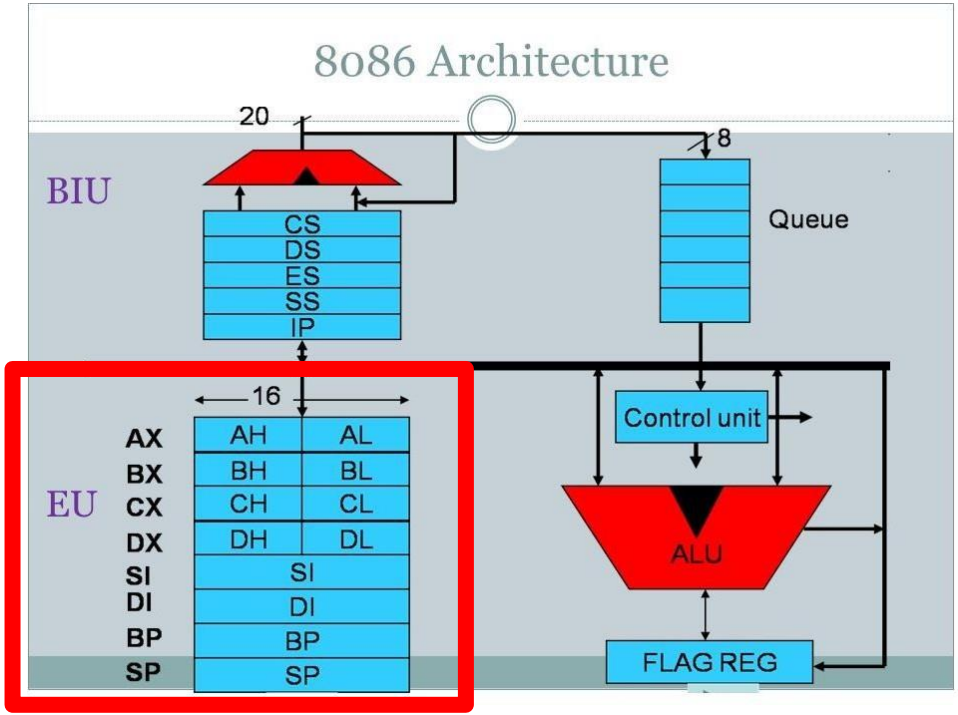
Used to hold base addresses for memory access. Helps in indirect addressing.

- **CX (Count Register)**

Acts as a loop counter in instructions like `LOOP`, `REP`, etc.

- **DX (Data Register)**

Holds data for I/O and used in multiplication/division with AX.



- **SI (Source Index)**

Points to source data in string operations. Works with DS segment.

- **DI (Destination Index)**

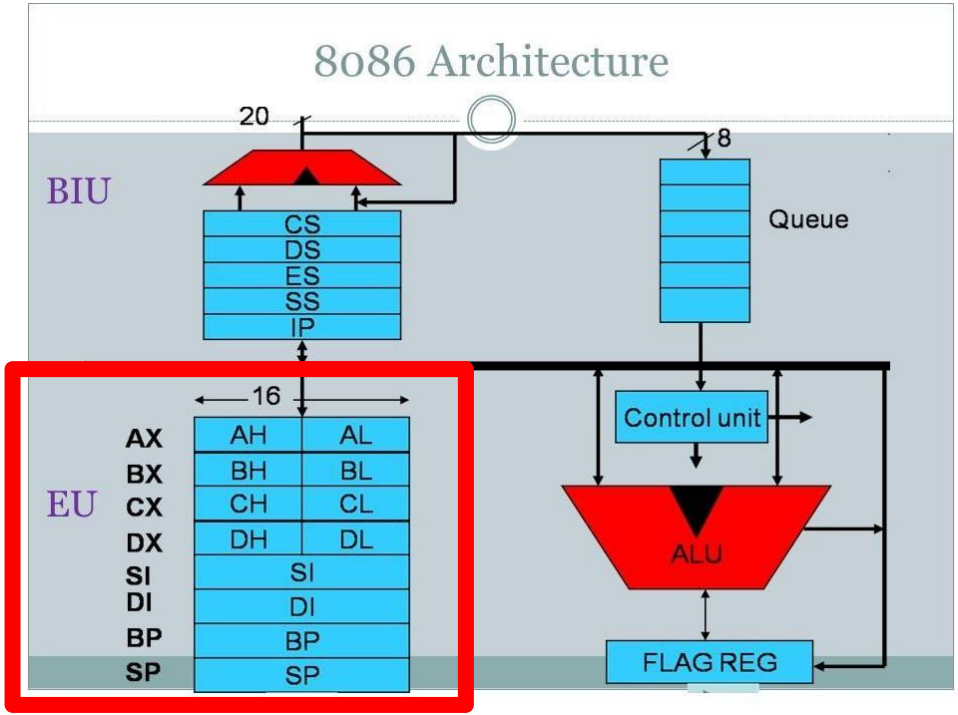
Points to destination in string operations. Works with ES segment.

- **SP (Stack Pointer)**

Points to the top of the stack. Used in `PUSH`, `POP`, `CALL`, and `RET`.

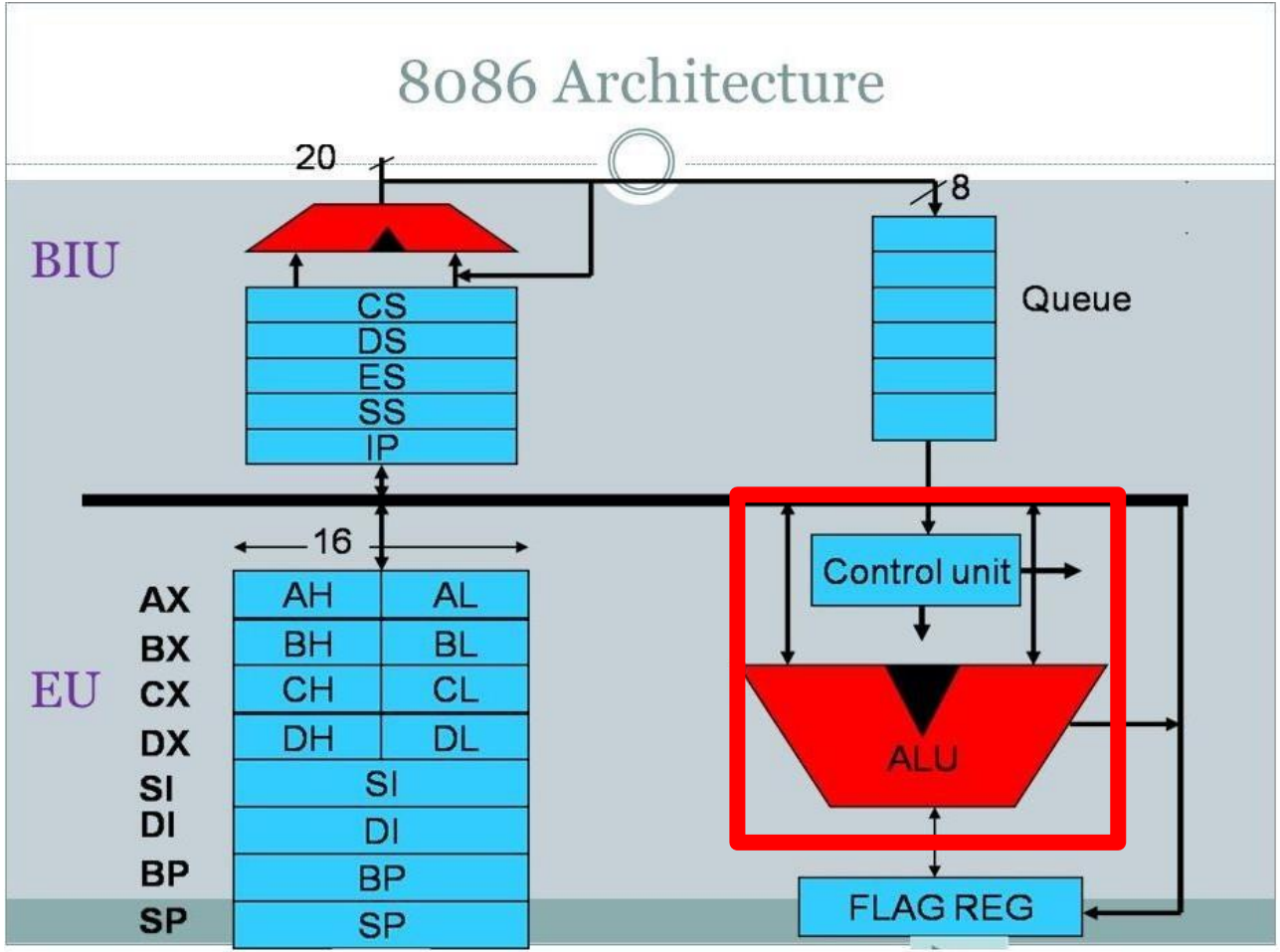
- **BP (Base Pointer)**

Accesses data in the stack frame. Used in function calls and local variable access.




Register Set
AH
AL
AL
AL
SP
BH
CH
DH
DI
SI

8086 Architecture



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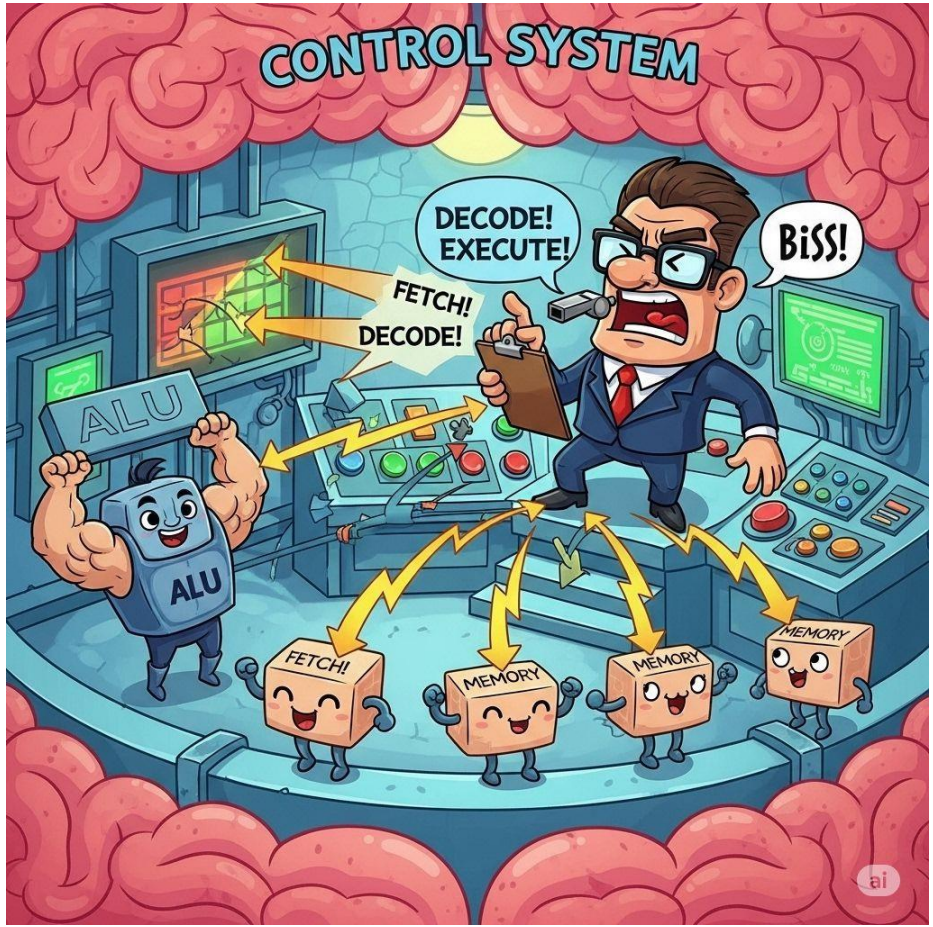
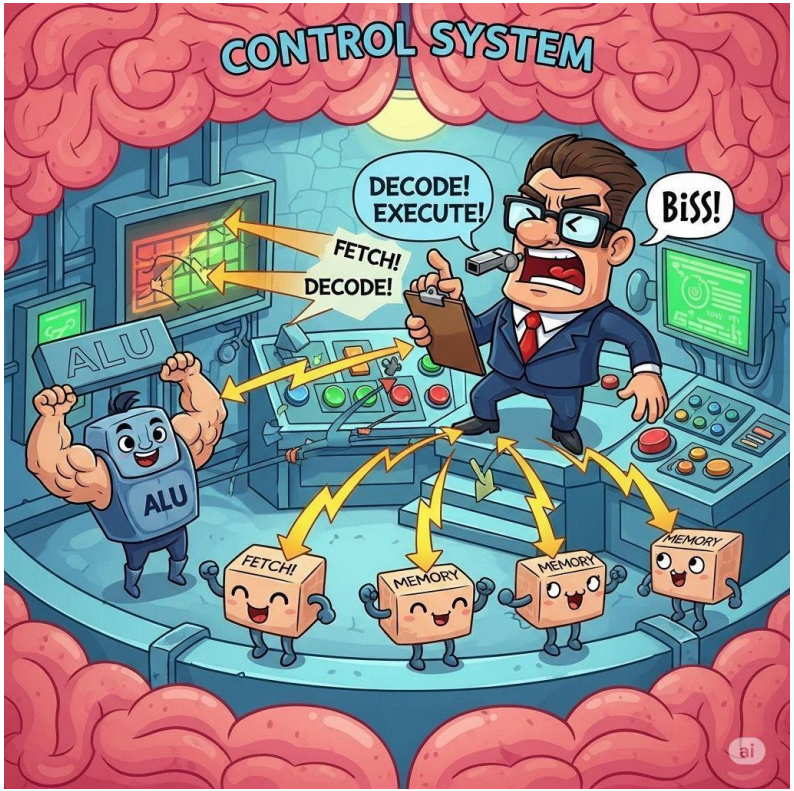




Control Unit (CU):

Definition:

The **Control Unit** manages and coordinates the execution of instructions by directing the operation of the ALU, registers, and memory.

Function in 8086:

- Decodes instructions from the **Instruction Queue**.
- Sends control signals to **ALU** and other components.
- Ensures proper sequencing of operations.

অঙ্কের মাস্তান ALU

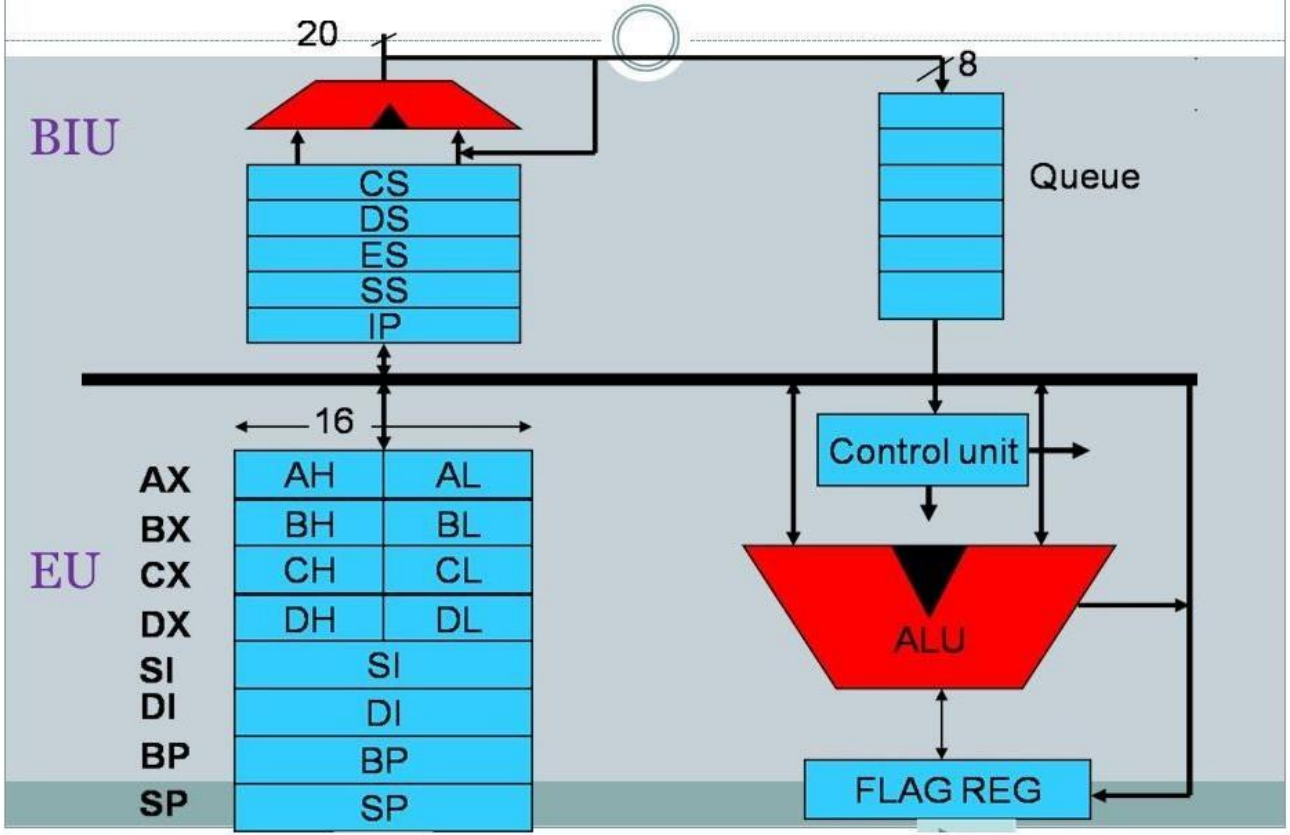
ALU (Arithmetic Logic Unit):

Definition:
The **ALU** performs all **arithmetic** (like addition, subtraction) and **logical** (like AND, OR, NOT) operations in the microprocessor.

Function in 8086:

- Executes operations based on instructions from the Control Unit.
- Affects **flags** in the **Flag Register** based on results (e.g., Zero, Carry).

8086 Architecture



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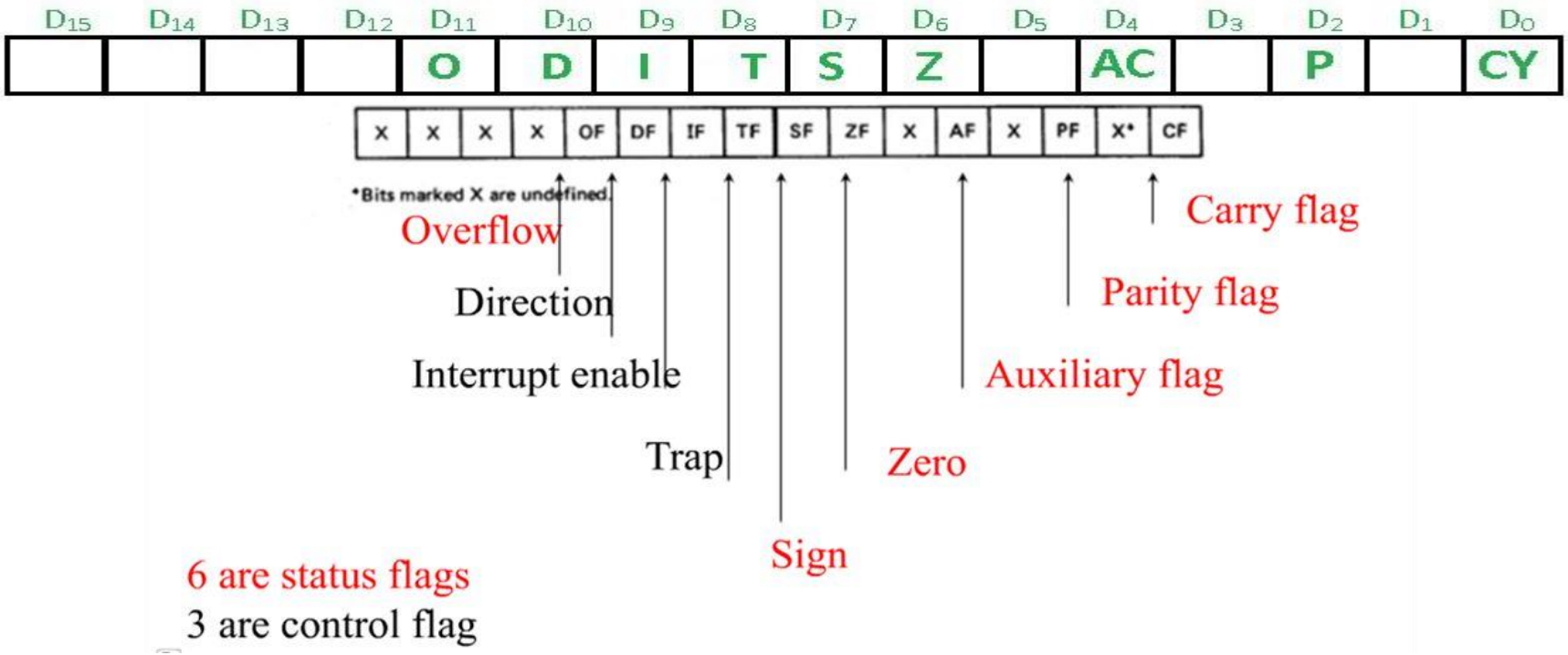
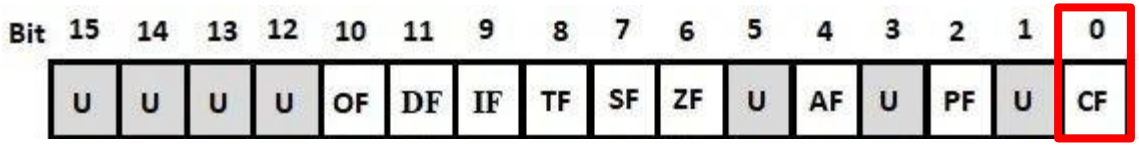
Flags



6 are status flags

3 are control flag



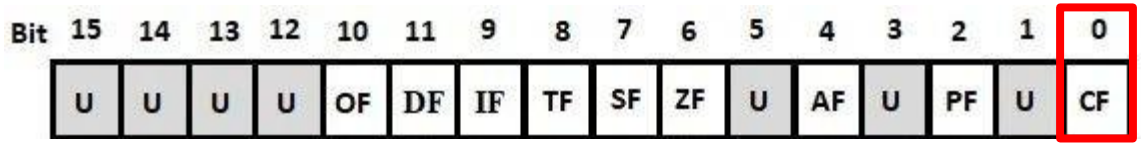
Carry Flag (CF)

- **Type:** Status Flag
- **Position:** D0
- **Definition:** Indicates an unsigned overflow, i.e., a carry out of the MSB in addition or a borrow in subtraction.
- **Affected by:** ADD, SUB, ADC, SBB, etc.

Example:

```
1111 1111 (255)
+
0000 0001 (1) = 1 0000 0000
            (Overflow) → CF = 1
```

- **Use Case:** Useful in multi-byte arithmetic or loop carry propagation.



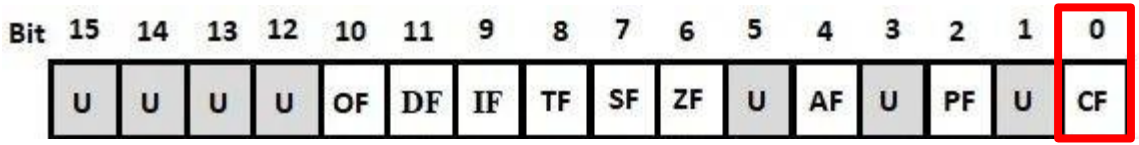
Parity Flag (PF)

- **Type:** Status Flag
- **Position:** D2
- **Definition:** Shows whether the number of 1s in the result is even.
- **Affected by:** Most arithmetic/logical operations

Example:

```
Result: 0000 1111 → Has four 1s → PF = 1 (Even)
Result: 0000 1011 → Has three 1s → PF = 0 (Odd)
```

- **Use Case:** Error checking or communication protocols.



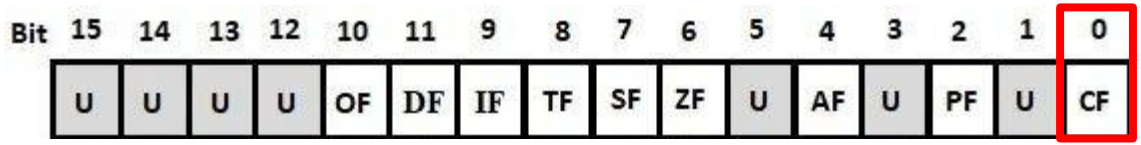
Auxiliary Carry Flag (AF)

- **Type:** Status Flag
- **Position:** D4
- **Definition:** Indicates a carry from bit 3 to bit 4. Used mainly in BCD operations.
- **Affected by:** ADD, SUB

Example:

```
1111 1111 (255)
+
0000 0001 (1) = 1 0000 0000
             AF = 1
```

- **Use Case:** Decimal (BCD) addition/correction.



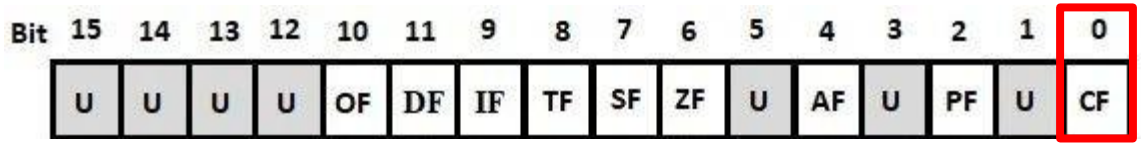
Zero Flag (ZF)

- **Type:** Status Flag
- **Position:** D6
- **Definition:** Indicates whether the result of an operation is zero.
- **Affected by:** CMP, SUB, AND, OR, XOR

Example:

```
0011 0000 - 0011 0000 = 0000 0000 → ZF = 1
```

- **Use Case:** Conditional execution, loops (e.g., JZ, JNZ).



Sign Flag (SF)

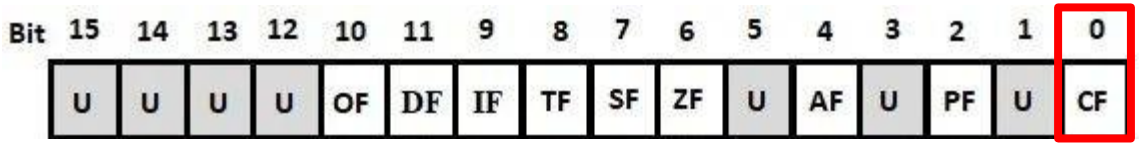
- **Type:** Status Flag
- **Position:** D7
- **Definition:** Reflects the sign (most significant bit) of the result.
- **Affected by:** Arithmetic and logical operations.

Example:

Result : `1000 0000 → SF = 1 (Negative Number)`

Result : `0100 1100 → SF = 0 (positive Number)`

- **Use Case:** Signed arithmetic and branching (e.g., JS, JNS).



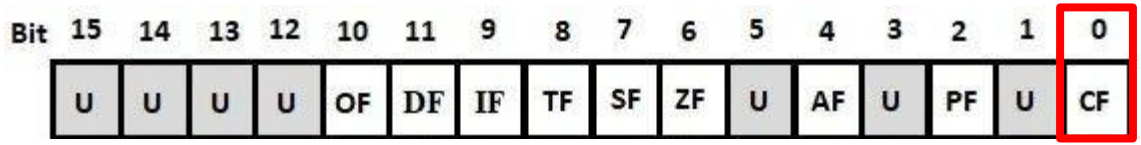
Trap Flag (TF)

- **Type:** Control Flag
- **Position:** D8
- **Definition:** Enables single-step debugging. When set, the processor interrupts after every instruction.

Example:

```
TF = 1 → Executes one instruction, then calls INT 1
```

- **Use Case:** Debuggers for instruction-by-instruction execution.



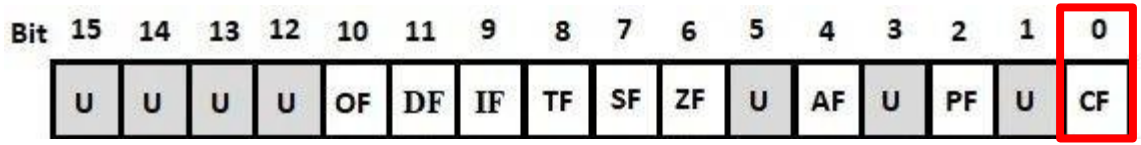
Interrupt Flag (IF)

- **Type:** Control Flag
- **Position:** D9
- **Definition:** Controls whether the processor will respond to maskable external interrupts.

Example:

```
IF = 1 → Interrupts are enabled (STI)
IF = 0 → Interrupts are disabled (CLI)
```

- **Use Case:** Critical code sections that shouldn't be interrupted.



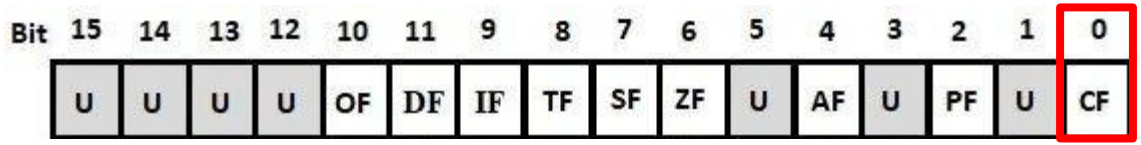
Direction Flag (DF)

- **Type:** Control Flag
- **Position:** D10
- **Definition:** Controls string processing direction:
  - `0` = Auto-increment (forward)
  - `1` = Auto-decrement (backward)

Example:

```
DF = 1 → Used in REP MOVSB to copy strings in reverse
```

- **Use Case:** `MOVS`, `CMPS`, `SCAS`, etc. for block operations.



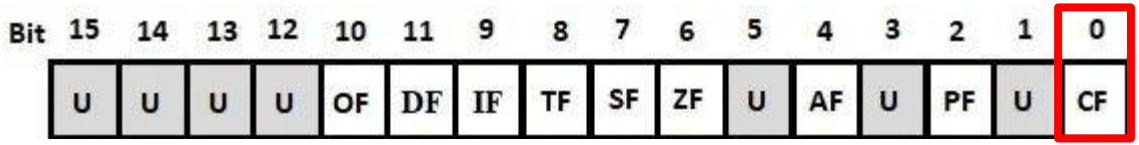
Overflow Flag (OF)

- **Type:** Status Flag
- **Position:** D11
- **Definition:** Indicates a signed overflow — when result is too large for destination in signed operations.

Example:

```
0111 1111 (+127)
+
0000 0001 (+1) = 1000 0000 → OF = 1
(Result looks like -128 in signed 8-bit!)
```

- **Use Case:** Signed comparison, conditional branching (JO, JNO)



❌ Unused/Reserved Bits (D1, D3, D5, D12–D15)

- These bits are reserved for internal or future use and should be ignored in 8086 programming.

🔹 Binary Addition Laws

- `0 + 0 = 0` (Carry 0)
- `0 + 1 = 1` (Carry 0)
- `1 + 0 = 1` (Carry 0)
- `1 + 1 = 0` (Carry 1)

📌 Carry is generated when both bits are 1.

🔹 Binary Subtraction Laws

- `0 - 0 = 0` (Borrow 0)
- `1 - 0 = 1` (Borrow 0)
- `1 - 1 = 0` (Borrow 0)
- `0 - 1 = 1` (Borrow 1)

📌 Borrow happens when subtracting 1 from 0.

Decimal **10** = Hexadecimal **A**

Decimal **11** = Hexadecimal **B**

Decimal **12** = Hexadecimal **C**

Decimal **13** = Hexadecimal **D**

Decimal **14** = Hexadecimal **E**

Decimal **15** = Hexadecimal **F**

Type	Decimal Range	Hex Range
Signed	**-128 to 127**	**0x80 to 0x7F**

Addition

AL = 25H = 0010 0101
ADD AL, 13H = 0001 0011
- - - - - - - - - - - - - -
Result = 0011 1000 = 38H

CF: 0

PF: 0

AF: 0

ZF: 0

SF: 0

OF: 0

AL = 7FH = 0111 1111
ADD AL, 01H = 0000 0001
- - - - - - - - - - - - - -
Result = 1000 0000 = 80H

CF: 0

PF: 0

AF: 1

ZF: 0

SF: 1

OF: 1 (signed: +127 + 1 → -128 → sign flipped)

AL = 50H = 0101 0000
ADD AL, 50H = 0101 0000
- - - - - - - - - - - - - -
Result = 1010 0000 = A0H

CF: 0

PF: 1 (even parity)

AF: 0

ZF: 0

SF: 1

OF: 1 (signed: +80 + 80 = -96, invalid sign change)

AL = F0H = 1111 0000
ADD AL, 20H = 0010 0000
- - - - - - - - - - - - - -
Result = 0001 0000 = 10H

CF: 1

PF: 0

AF: 0

ZF: 0

SF: 0

OF: 0

Subtraction

AL = A0H = 1010 0000
SUB AL, 20H = 0010 0000

Result = 1000 0000 = 80H

CF: 0

PF: 0

AF: 0

ZF: 0

SF: 1

OF: 0

AL = 7FH = 0111 1111
SUB AL, FFH = 1111 1111

Result = 1000 0000 = 80H

CF: 1

PF: 0

AF: 1

ZF: 0

SF: 1

OF: 1 (signed: 127 - (-1) = 128 → overflow)

SUB AL, 10H

```
AL      = 3CH = 0011 1100
Operand = 10H = 0001 0000
---------------------------
Result  = 2CH = 0010 1100
```

🏳 **Flag Status:**

- **CF = 0**.
- **PF = 0**
- **AF = 0**
- **ZF = 0**
- **SF = 0**
- **OF = 0** → No signed overflow (positive - positive = positive).

AL = 80H = 1000 0000
SUB AL, 01H = 0000 0001
- - - - - - - - - - - - - -
Result = 0111 1111 = 7FH

CF: 0

PF: 0 (7 ones = odd → PF = 0)

AF: 1

ZF: 0

SF: 0

OF: 1 (signed: -128 – 1 = +127 → overflow)